

Table 1. Paper and Methodology Information

SL	Paper Title	Authors	Year	FACED?	Dataset	Application Area	Input Features	Graph Construction	Method/Model	Evaluation Strategy
1	EEG Emotion Recognition Using Dynamical Graph Convolutional Neural Networks	Tengfei Song; Wenming Zheng; Peng Song; Zhen Cui	2020	No	SEED: 15 subjects, 62 channels, 3 emotions; DREAMER: 23 subjects, 14 channels, binary valence/arousal/dominance	EEG emotion recognition and functional channel-relation learning	SEED: DE, PSD, DASM, RASM, DCAU; DREAMER: PSD	EEG channels as nodes; one trainable shared adjacency matrix	DGCNN	Subject-dependent and subject-independent/LOSO experiments; DREAMER evaluation under the authors' session-independent setting
2	EEG-Based Emotion Recognition Using Regularized Graph Neural Networks	Peixiang Zhong; Di Wang; Chunyan Miao	2022	No	SEED and SEED-IV: 15 subjects, 62 channels; 3 and 4 emotion classes	Cross-subject EEG emotion recognition	Multi-band differential entropy	Local distance edges, inter-hemispheric edges and learnable sparse symmetric adjacency	RGNN with NodeDAT and EmotionDL regularization	Subject-dependent and subject-independent experiments with ablation analysis
3	Cross-Subject EEG Emotion Recognition With Self-Organized Graph Neural Network	Jingcong Li; Shuqi Li; Jiahui Pan; Fei Wang	2021	No	SEED and SEED-IV: 15 subjects, 62 channels	Cross-subject emotion recognition	Five-band DE extracted from temporal segments	Sample-specific adjacency generated through softmax weighting and top-k sparsification	SOGNN with temporal convolution, self-organized graph learning and GCN	Leave-one-subject-out cross-validation
4	A Large Finer-Grained Affective Computing EEG Dataset	Jingjing Chen; Xiaobin Wang; Chen Huang; Xin Hu; Xinke Shen; Dan Zhang	2023	Yes	FACED: 123 subjects, 32 channels, 28 video stimuli and 9 emotion categories	Fine-grained EEG affective computing and dataset development	Raw/preprocessed EEG, five-band DE and PSD	No graph construction	Dataset introduction with machine-learning and contrastive-learning baselines	Intra-subject and cross-subject baseline evaluation
5	LGGNet: Learning From Local-Global-Graph Representations for Brain-Computer Interface	Yi Ding; Neethu Robinson; Chengxuan Tong; Qiuhaio Zeng; Cuntai Guan	2024	No	Three BCI datasets, including DEAP: 32 subjects and 32 channels	Emotion, preference, attention and fatigue classification	Multi-scale temporal convolution features	Predefined local brain-region graphs and learned global inter-region graph	LGGNet	Nested cross-validation, ablation, statistical testing and visualization
6	GNN4EEG: A Benchmark and Toolkit for Electroencephalography Classification With Graph Neural Network	Kaiyuan Zhang; Ziyi Ye; Qingyao Ai; Xiaohui Xie; Yiqun Liu	2024	Yes	FACED: 123 subjects; 32 recorded channels, with 30 channels used after preprocessing; 2-, 3-, 4- and 9-class tasks	Standardized benchmarking of EEG-GNN models	Five-band DE with LDS smoothing	Channels as nodes; DGCNN, RGNN, SparseDGCNN and HetEmotionNet graph settings	GNN4EEG benchmark and toolkit	Intra-subject/cross-subject splits with FCV, NCV and nested-CV protocols
7	EEG-Based Emotion Recognition Using Graph Convolutional Neural Network With Dual Attention Mechanism	Wei Chen; Yuan Liao; Rui Dai; Yuanlin Dong; Liya Huang	2024	No	DEAP: 32 subjects, 32 channels; SEED and SEED-IV: 15 subjects, 62 channels	EEG emotion recognition and attention-based interpretation	Four- or five-band DE	Fixed adjacency based on three-dimensional electrode distances	DAMGCN with residual GCN, channel attention and frequency attention	Ten-fold CV, leave-one-trial-out, LOSO and ablation experiments
8	Dynamic-Attention-Based EEG State Transition Modeling for Emotion Recognition	Xinke Shen; Runmin Gan; Kaixuan Wang; Shuyi Yang; Qingzhu Zhang; Quanying Liu; Dan Zhang; Sen Song	2025	Yes	FACED, SEED and SEED-V	Cross-subject EEG emotion decoding and state-transition modelling	Temporal and spatial EEG transition representations	No explicit channel graph; dynamic temporal and spatial state components	DAEST with dynamic attention and contrastive learning	Cross-subject evaluation on binary and multiclass tasks

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9	Cross-Subject EEG Signals-Based Emotion Recognition Using Contrastive Learning	Ahmed Mohammed Alghamdi; M. Usman Ashraf; Adel A. Bahaddad; Khalid Ali Almarhabi; Waleed A. Al Shehri; Amil Daraz	2025	Yes	FACED, SEED, CEED and MPED	Cross-subject EEG emotion recognition	Spatial, temporal and frequency-domain representations	No graph; region-aware representation learning	CSCL with dual contrastive objectives, hyperbolic representation and domain adaptation	Cross-subject evaluation across multiple datasets
10	EmT: A Novel Transformer for Generalized Cross-Subject EEG Emotion Recognition	Yi Ding; Chengxuan Tong; Shuailei Zhang; Muyun Jiang; Yong Li; Kevin Lim Jun Liang; Cuntai Guan	2025	Yes	FACED, SEED and additional emotion-classification datasets; MAHNOB-HCI for regression	Generalized cross-subject emotion classification and regression	Seven-band relative PSD from short- and long-term segments	Temporal sequence of EEG feature graphs with multiple learnable graph views	EmT: temporal graph construction, residual multi-view pyramid GCN and temporal Transformer	Generalized cross-subject classification and regression experiments
11	MSGM: A Multi-Scale Spatiotemporal Graph Mamba for EEG Emotion Recognition	Hanwen Liu; Yifeng Gong; Zuwei Yan; Zeheng Zhuang; Jiaxuan Lu	2026	Yes	FACED, SEED and THU-EP	Subject-independent EEG emotion recognition	Seven-band relative PSD with multiple temporal-window scales	Global-local adaptive graphs guided by neuroanatomical brain-region priors	MSGM with multi-depth GCN, token fusion and Mamba	Subject-independent evaluation, including leave-n-subject-out settings
12	Stimulus Identity Rather Than Emotion Drives EEG Classification on the FACED Dataset	Moritz Gerster; Elizaveta Sirotina; Anton Orlovskii; Alexandra Hertz; Juliette Champaud; Domenico Guarino; Silvia Tulli	2026	Yes	FACED: 123 subjects, 32 channels, 28 videos and 9 emotions	Confound analysis and validity assessment of FACED emotion classification	Official FACED DE features	No graph construction	LinearSVC and CLISA-based analyses	Intra-/cross-subject, self-report-aware and reduced-stimulus experiments

Table 2. Results and Critical Analysis

SL	Metrics / Key Results	Strengths	Limitations	Research Gap	Relation to Our Work
1	SEED: 90.40% subject-dependent, 79.95% LOSO; DREAMER: 86.23% V, 84.54% A, 85.02% D	Learns functional channel relationships end-to-end	Uses one shared graph; limited temporal modelling	Sample-specific, temporally adaptive graphs are needed	Core GNN baseline for adaptive graph and temporal improvements
2	Outperformed several baselines on SEED and SEED-IV	Combines biological priors, sparse graphs and domain adaptation	Relies partly on manual priors; weak temporal modelling	Adaptive yet interpretable temporal graphs are needed	Supports regularized and interpretable FACED-9 graph learning
3	SEED: 86.81%; SEED-IV: 75.27%	Learns a separate graph for each sample	Higher complexity; small datasets; limited neurological constraints	Dynamic graphs need validation on larger datasets	Motivates sample-specific graph learning for FACED
4	Demonstrated intra- and cross-subject FACED-9 classification	Large, fine-grained and diverse EEG dataset	Dataset paper; difficult cross-subject task; possible stimulus confounding	Robust and stimulus-aware FACED-9 models are needed	Provides the target dataset, features and baseline protocols
5	DEAP emotion: 64.51% F1; preference: 63.07% accuracy, 72.53% F1	Combines temporal, local and global graph features	Uses manually defined regions; not designed for FACED-9	More adaptive local-global graphs are needed	Inspires local-global modelling with fewer fixed priors
6	NCV: Cross-9 RGNN 31.3%; Intra-9 HetEmotionNet 44.9%	Provides standardized GNN evaluation protocols	Benchmark only; performance varies by validation protocol	Stronger leakage-resistant FACED-9 models are needed	Supplies fair baselines and evaluation settings
7	Subject-dependent: 99.42% SEED, 96.86% SEED-IV; trial-independent: 78.21% DEAP, 90.40% SEED, 83.55% SEED-IV	Adds channel and frequency attention	Fixed distance graph may miss functional connectivity	Spatial and learned functional graphs should be combined	Supports attention with adaptive graph construction
8	FACED-2: 75.4%; FACED-9: 59.3%; SEED: 88.1%; SEED-V: 73.6%	Strong dynamic state and temporal modelling	Not graph-based; limited channel-level interpretation	Dynamic temporal learning should be combined with graphs	Strong non-GNN FACED baseline for temporal modelling
9	FACED: 65.98% accuracy	Learns subject-invariant representations	No channel graph; protocol-dependent comparison; possible confounding	Contrastive learning should include graph and stimulus controls	Motivates contrastive learning within the proposed GNN
10	FACED: 60.8% accuracy, 74.0% F1; SEED: 80.2% accuracy, 82.1% F1	Combines multi-view graphs and long-range temporal modelling	Complex and difficult to interpret	Simpler, interpretable graph-temporal models are needed	Closest reference for graph and Transformer integration
11	FACED: 63.17% accuracy, 76.01% F1	Models multi-scale temporal and local-global connectivity	Uses predefined regions; stimulus confounding not tested	More adaptive graphs and stimulus-controlled validation are needed	Strong recent FACED baseline for graph-temporal comparison
12	Strong performance remained despite self-report mismatch; reduced-stimulus tests also performed well	Directly examines stimulus-identity confounding	Preprint; not GNN-based; needs further validation	Stimulus-independent and self-report-aware evaluation is needed	Defines the main validity test for the proposed model